

# Synthetic Industrial Surface Defect Generation Under Low-Data Constraints

Evaluating Conditional GANs for Defect-Faithful Image Generation from Limited Samples

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## Abstract

Industrial defect datasets are often small, highly imbalanced, expensive to collect, and difficult to share, creating a major bottleneck for computer vision systems used in automated inspection. This challenge is especially pronounced in metal-surface imagery, where rare defect cues must be learned against dominant background texture. This report investigates whether conditional generative models can synthesize useful, defect-faithful industrial surface images under such low-data constraints. The study focuses on a grayscale metal-surface dataset with three classes—normal, scratch, and spot—and uses a conditional Deep Convolutional Generative Adversarial Network (cDCGAN) as a simple label-conditioned baseline. A staged methodology is developed: a three-class baseline is first evaluated, then decomposed into normal-vs-scratch and normal-vs-spot tracks, followed by targeted low-data interventions. Two controlled follow-up experiments then test whether the scratch-generation limitations can be explained primarily by complex metal texture or insufficient scratch exposure. A procedurally generated simple-scratch dataset removes the metal background and systematically varies scratch geometry. Under balanced training, the model occasionally produces small scratch-like cues but does not learn reliable class consistency or geometric diversity. A scratch-heavy ablation increases scratch exposure from 50% to 87.5% of each batch, but produces similar point- and grid-like artifacts under both labels and degrades normal-class fidelity. These findings show that background complexity and class imbalance contributed to the original difficulty but do not fully explain it. The present label-conditioned cDCGAN and training objective are insufficient for reliable, controllable scratch synthesis, motivating future work in spatial guidance, explicit geometric conditioning, defect transfer, and image-to-image translation. **Keywords:** Generative Adversarial Networks, Conditional GANs, Synthetic Defect Generation, Industrial Computer Vision, Low-Data Learning, Surface Defect Detection, Data Scarcity.

## 1 Introduction

In industrial inspection settings, surface defect datasets are often small, imbalanced, expensive to annotate, and difficult to acquire under realistic operating conditions [3]. These constraints create a major bottleneck for computer vision systems used in automated inspection, particularly in manufacturing domains where rare but safety-relevant defects must be detected reliably. The challenge is especially pronounced in metal-surface imagery, where the visual signal of interest may occupy only a small portion of the image while the majority of pixels represent shared background texture.

This work addresses the data side of that inspection problem. Rather than directly training a downstream detector, the study investigates whether conditional generative models can produce synthetic industrial defect images that are realistic enough, defect-faithful enough, and varied enough to eventually support inspection-model training under low-data constraints. The objective is not simply to generate visually plausible images, but to understand what kinds of defect structures are learnable when real defect data is scarce and imbalanced, which interventions improve generation quality, and which limitations remain unresolved.

Among generative approaches, conditional Generative Adversarial Networks (cGANs) provide a natural starting point because they allow image generation to be guided by a requested class label [4]. This makes them well suited for testing whether a model can respond to a requested industrial defect type, such as a scratch or a spot, within a constrained data regime. At the same time, such models are not expected to provide full spatial control over defect placement or geometry, making them useful as a baseline for identifying both learnable patterns and architectural limitations.

The study is organized around four research questions:

1. Can the model generate the requested defect type?
2. Which defect types are easier to learn?

3. What improves output quality when real defect data is scarce and imbalanced?
4. Do scratch-generation failures persist after complex background texture and unequal scratch exposure are removed as competing explanations?

To answer these questions, the work begins with a label-conditioned cDCGAN baseline trained across three classes—normal, scratch, and spot. After observing limitations in the three-class setting, the task is simplified through binary decomposition into normal-vs-scratch and normal-vs-spot tracks. Several low-data interventions are then evaluated, including oversampling, stratified batch sampling, defect-specific enhancement, and mild geometric augmentation. Finally, two controlled simple-scratch experiments isolate the effects of background complexity and scratch exposure while preserving the same architecture and training logic. The resulting contribution is not a production-ready defect generator, but a structured experimental framework for identifying which limitations arise from the data and which persist under simplified conditions, thereby providing a defensible basis for selecting more controllable future methods.

## 2 Related Work

Generative models have become an important tool for synthetic data creation when labeled samples are scarce. In computer vision, Generative Adversarial Networks (GANs) introduced adversarial learning as a way to synthesize images from latent noise by training a generator and discriminator in competition [1]. Conditional GANs (cGANs) extended this framework by incorporating side information such as class labels, allowing generation to be guided toward a requested category [4]. For image synthesis tasks, convolutional GAN variants such as DCGAN further demonstrated that deep convolutional architectures can improve training stability and visual quality relative to simpler fully connected designs [5]. Within the context of this study, a label-conditioned cDCGAN provides a natural baseline for asking whether a model can generate specific industrial defect classes under low-data constraints.

Beyond noise-to-image generation, image-to-image translation methods provide an alternative paradigm for synthetic defect creation by transforming an existing source image into a target-domain image. Supervised approaches such as pix2pix learn paired mappings between source and target images [2], while CycleGAN enables unpaired translation through cycle-consistency constraints [7]. These methods are highly relevant to industrial defect synthesis because they can preserve realistic background texture while focusing on defect insertion or transformation. However, they also introduce practical trade-offs. Paired clean-to-defect image datasets are difficult to obtain in industrial settings, making supervised translation

less realistic in many applications, while unpaired translation often lacks precise control over defect placement, geometry, and severity.

More recent defect-specific synthesis work suggests that stronger controllability may require transfer-based or spatially guided strategies rather than global class conditioning alone. Defect Transfer GAN, for example, was proposed to synthesize diverse industrial defects for data augmentation and is directly relevant both as a defect-specific synthesis method and as the source of the SDI dataset family used in this project [6]. Such approaches motivate the future direction of the present work: moving beyond simple label-conditioned generation toward methods that better preserve surface realism while providing more explicit control over defect structure. In this sense, the current cDCGAN experiments are positioned not as a final solution, but as a structured baseline for identifying what defect cues are learnable under scarce-data conditions and where more controllable architectures become necessary.

## 3 Data

The experiments use a grayscale industrial metal-surface dataset derived from the Surface Defect Inspection (SDI) dataset family, specifically a Type-A metal-surface subset [6]. The study intentionally restricts the experiments to a single surface family so that the model is not simultaneously required to learn multiple material domains. This keeps the background surface distribution relatively consistent and makes defect generation the central challenge rather than cross-domain texture variation.

The full SDI dataset contains multiple surface types and a substantially larger number of images, but the present work focuses on the Type-A subset and adopts a class distribution aligned with the baseline image-synthesis setup reported in the Defect Transfer GAN work [6]. The selected counts are:

- Normal: 702 images
- Scratch: 344 images
- Spot: 181 images

This subset choice was motivated not only by consistency with prior work, but also by the desire to preserve a realistic low-data industrial setting. In many practical inspection scenarios, it is difficult to obtain large numbers of diverse defect samples, especially for rare classes. Restricting the study to a few hundred images per class therefore provides a more realistic test of whether class-conditioned generative models can learn useful defect cues under scarcity. At the same time, the resulting class imbalance creates an additional challenge: matching minority defect classes to the larger normal class through repetition-based oversampling can improve class exposure during training, but it can also introduce bias by repeatedly presenting the same rare samples.

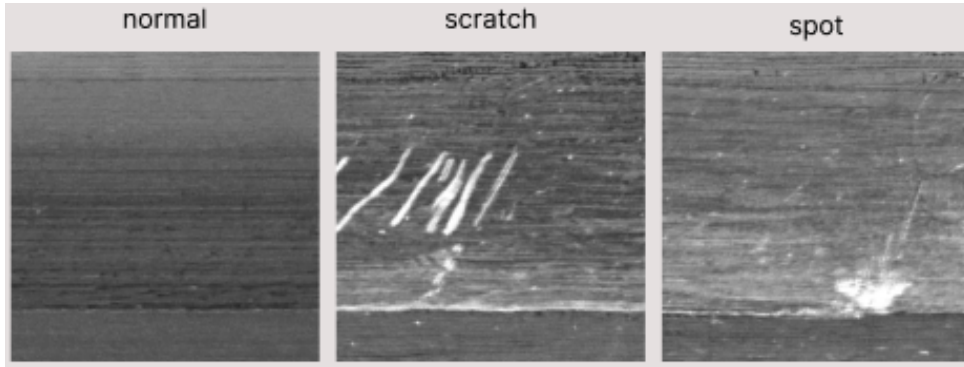


Figure 1: Representative real images from the Type-A subset of the SDI dataset used in this study. From left to right: normal, scratch, and spot. The examples illustrate both the shared metal background texture and the relatively small visual footprint of the defect signal.

All images are resized to  $128 \times 128$  and represented in grayscale. This image size was chosen to reduce computational load while preserving the primary visible defect cues. Even under this simplification, the setting remains difficult because only a relatively small portion of each image corresponds to the defect itself, while most pixels describe shared metal texture. In other words, the experiments operate in a high background-texture / low defect-signal regime.

The class counts further highlight the central low-data constraint of the project. The dataset is not only limited in size but also strongly imbalanced, especially for the spot class. This imbalance directly motivated the experimental design and the low-data interventions introduced later in the study, including oversampling, stratified batch sampling, and defect-specific enhancement.

## 4 Methods

### 4.1 Baseline Model Choice

The study begins with a conditional Deep Convolutional Generative Adversarial Network (cDCGAN) as a simple label-conditioned baseline. This model was selected to test whether a generator could respond to a requested defect class under scarce-data conditions. The conditioning signal is a class label  $y \in \{\text{normal}, \text{scratch}, \text{spot}\}$ .

Conditional GANs are a useful starting point for this problem because they allow generation to be guided by a requested category while retaining a relatively simple adversarial training framework. In the present context, this makes it possible to test whether the model can generate class-specific industrial defect imagery from limited and imbalanced data, even if full spatial control is not yet available.

### 4.2 Model Overview

Conceptually, the generator receives a latent noise vector  $z$  together with a class label  $y$  and produces a synthetic

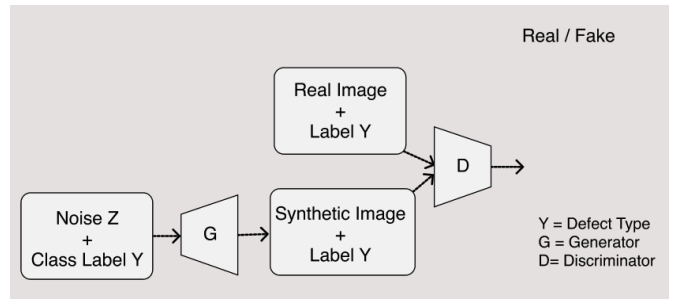


Figure 2: Simplified cDCGAN framework used in this study. The generator receives latent noise  $z$  and class label  $y$  to produce a synthetic image, while the discriminator evaluates image-label pairs as real or fake.

grayscale image. The latent input  $z$  serves as the stochastic source signal for image generation, while the class label specifies the requested defect type. The discriminator then receives either a real image with its label or a generated image with its label and predicts whether the image-label pair is real or fake. This setup makes class consistency an explicit part of the learning problem.

A simplified overview of the conditional generation process is shown in Figure 2. The figure is intended to communicate the conditioning logic of the model rather than the full layer-by-layer architecture.

### 4.3 Experimental Setup

The initial experiment trains a single three-class cDCGAN to generate normal, scratch, and spot images within one shared model. This serves as the baseline and reveals whether a single conditional model can learn both the shared metal texture and the class-specific defect structures.

After limitations are observed in the three-class setting, the problem is simplified through binary decomposition. Two separate models are trained:

- Normal vs Scratch

- Normal vs Spot

The purpose of this decomposition is to reduce conditional complexity and compare defect-specific learnability more directly. It also allows defect-type-specific interventions to be tested in a more controlled way.

#### 4.4 Low-Data Training Strategies

To improve performance under scarcity and imbalance, several targeted training strategies are evaluated:

- **Oversampling with repetition** to reduce the effective imbalance between defect classes.
- **Stratified batch sampling** to make class exposure more stable during training.
- **Randomized defect-specific enhancement** to emphasize defect cues in the scratch and spot settings.
- **Mild geometric augmentation** to introduce limited variation without destroying the primary defect structure.

These interventions are not treated as generic preprocessing steps; instead, they are part of the low-data methodology being evaluated. The goal is to determine whether targeted changes in data exposure and defect visibility improve generation quality under constrained training conditions.

#### 4.5 Qualitative Evaluation Criteria

Generated outputs are evaluated using three qualitative criteria:

- **Visual realism:** whether the generated image plausibly matches the metal-surface domain.
- **Class consistency:** whether the generated image reflects the requested defect label.
- **Diversity:** whether outputs within the same class vary meaningfully rather than repeating similar visual structures.

These criteria are used throughout the experiments to compare the three-class baseline, the binary decomposition tracks, and the effects of the low-data interventions.

#### 4.6 Implementation and Training Infrastructure

The training pipeline is implemented in TensorFlow/Keras and executed on the RIT Research Computing cluster. The main training environment used one NVIDIA A100 GPU, 64 GB memory, and 8 CPUs per task. Early runs encountered out-of-memory issues, which led to a refactor of the training workflow using `tf.data` and `GradientTape`. This refactor improved training stability, reduced memory pressure, and made controlled comparisons across experiments feasible.

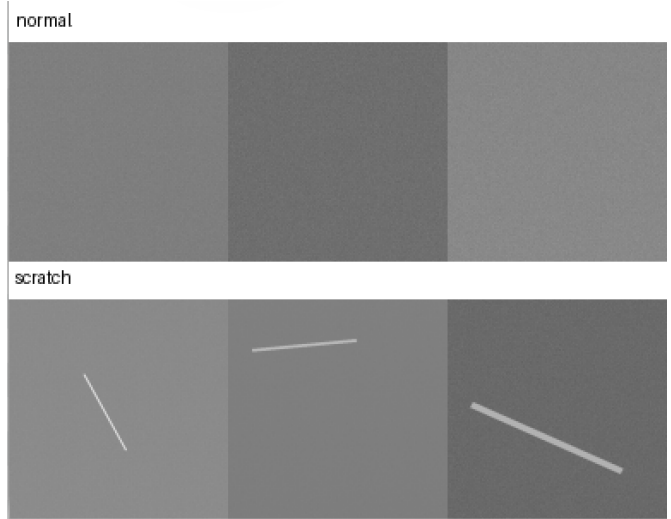


Figure 3: Representative samples from the controlled procedural dataset. Normal images contain only mild grayscale background variation. Scratch images use the same background process with one added line-like defect whose angle, position, length, width, and intensity vary.

#### 4.7 Controlled Simple-Scratch Follow-Up

The original experiments suggested that the cDCGAN may have devoted more capacity to the dominant metal background than to the relatively sparse scratch signal. To test that interpretation directly, a controlled procedural dataset was created in which the metal texture was removed. All images remained  $128 \times 128$  and grayscale. Normal images contained a mildly varying gray background with low-amplitude noise, while scratch images used the same background process with one added bright line-like structure. Scratch angle, position, length, width, and intensity were varied during dataset generation, and the associated parameters were stored as metadata. This design made the intended geometry known and removed real-image enhancement, geometric augmentation, and repetition-based oversampling as competing factors.

**Experiment 7: balanced controlled training.** The first controlled experiment used 700 unique normal images and 700 unique scratch images. Each batch contained eight normal and eight scratch samples. The existing binary cDCGAN architecture, latent dimension of 100, optimizer settings, `tf.data` pipeline, and `GradientTape` training loop were retained. Training was run for 82 epochs.

**Experiment 8: scratch-heavy exposure.** A second ablation tested whether the balanced setting gave too much relative exposure to the easier normal class. The experiment retained all 700 scratch images but used only 100 normal images. Each batch contained two normal and fourteen scratch samples, so all 100 normal and 700 scratch images were presented once per epoch across 50 batches. All other model and training settings were held



constant.

## 4.8 Fixed-Latent Checkpoint Evaluation

A fixed set of latent vectors was generated before training and reused at every saved checkpoint. For each latent vector, one normal-conditioned and one scratch-conditioned output were produced. Because the paired images share the same latent input and differ only in the class label, this evaluation makes it possible to inspect whether the conditioning label changes the generated structure. Separate grids using newly sampled latent vectors were also saved to inspect broader output diversity.

## 5 Results

Generated outputs were evaluated using the qualitative criteria defined in the Methods section: visual realism, class consistency, and diversity. Since this phase focused on understanding model behavior rather than validating downstream task performance, evaluation was based primarily on visual inspection of representative generated outputs and checkpoint progression.

The original metal-surface experiments revealed four main findings.

**1. The three-class cDCGAN learned shared surface texture more readily than defect-specific structure.** In the baseline setting, the generated outputs often resembled the overall metal-surface domain, but the scratch and spot classes were not consistently distinguishable. This suggests that the shared background texture was easier for the model to learn than the relatively small and localized class-defining defect cues.

**2. Binary decomposition improved class separation.** When the problem was simplified into separate normal-vs-scratch and normal-vs-spot tracks, the generated outputs showed clearer separation between normal and defect classes. Reducing the conditional complexity appeared to help the model focus more effectively on defect-specific structure.

**3. Defect geometry affected learnability.** The spot class was easier to model than the scratch class. Localized spot-like defects were learned more reliably, whereas scratches remained more difficult because they require coherent spatial structure, orientation, and placement across the image. Even when scratch outputs became more class-consistent, they often remained structurally weak relative to real examples.

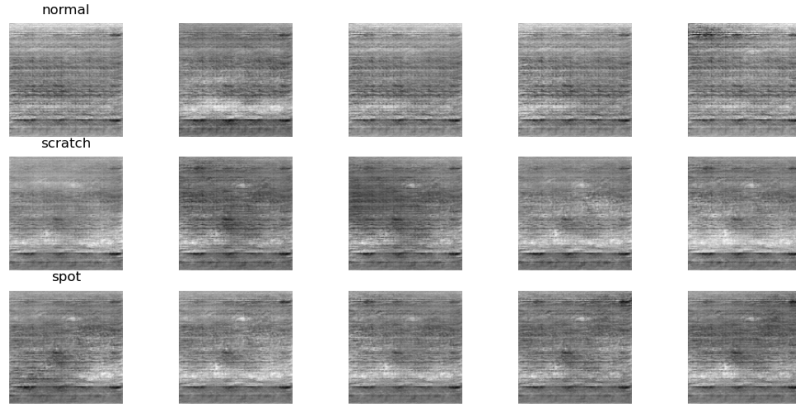
**4. Targeted low-data interventions improved class consistency more than diversity.** Oversampling, stratified batch sampling, defect-specific enhancement, and mild augmentation improved the visibility of

| Track             | Realism | Consistency | Diversity |
|-------------------|---------|-------------|-----------|
| 3-class baseline  | Medium  | Low         | Low       |
| Normal vs Scratch | Medium  | Medium      | Low       |
| Normal vs Spot    | Medium  | High        | Low       |

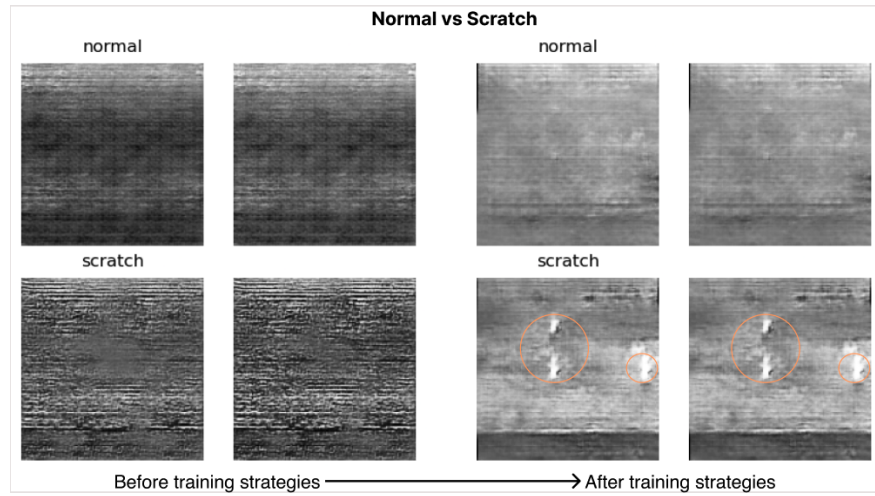
Table 1: Qualitative evaluation summary for the original metal-surface experiments.

class cues, especially in the binary settings. The clearest gains were observed for the spot track. However, diversity remained limited, and outputs frequently repeated similar visual patterns.

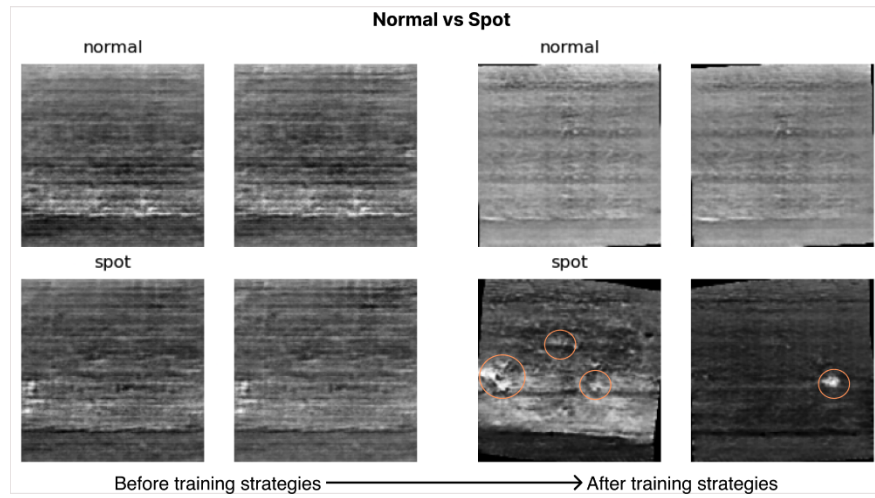
Table 1 summarizes the qualitative outcomes of the original metal-surface experiments, while Figure 4 shows the corresponding visual progression.



(a) Three-class cDCGAN baseline. Five generated samples are shown for each class. The model learned shared metal-surface texture more readily than clear class-specific defect structure, with limited within-class diversity.



(b) Normal-vs-scratch progression before and after targeted low-data interventions. Class separation improved, but scratch structure remained limited and repetitive.



(c) Normal-vs-spot progression before and after targeted low-data interventions. The spot class showed the clearest gains in class consistency, although output diversity remained limited.

Figure 4: Visual progression across the original experiment tracks. Top: three-class baseline outputs. Middle: normal-vs-scratch progression. Bottom: normal-vs-spot progression. Binary decomposition and targeted low-data strategies improved class consistency more clearly than diversity.

| Experiment         | Images    | Batch  | Separation | Geometry |
|--------------------|-----------|--------|------------|----------|
| Exp7 balanced      | 700N/700S | 8N/8S  | Weak       | Weak     |
| Exp8 scratch-heavy | 100N/700S | 2N/14S | Weak       | Weak     |

Table 2: Qualitative comparison of the controlled simple-scratch experiments. N denotes normal and S denotes scratch.

## 5.1 Controlled Simple-Scratch Results

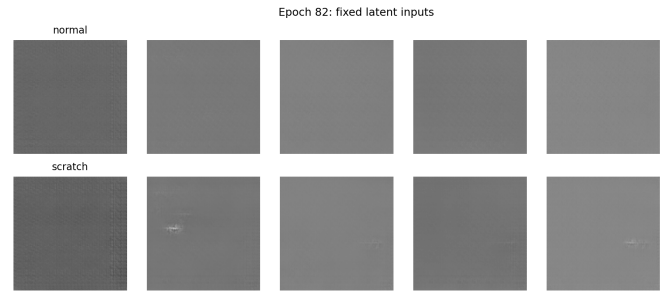
The two controlled experiments isolate different explanations for the earlier scratch-generation limitations. Table 2 summarizes the experimental contrast and qualitative outcomes.

**Experiment 7: balanced controlled training.** Removing the complex metal texture was not sufficient for reliable scratch generation. Balanced training on the procedural dataset produced cleaner normal backgrounds and occasional localized scratch-like cues. However, class consistency remained weak: at the final fixed-latent checkpoint, only two of the five scratch-conditioned outputs contained clearly visible defect-like structures. The few visible structures were small, mostly horizontal, and did not reproduce the intended variation in angle, position, length, and width. The original metal texture was therefore not the only obstacle to scratch generation.

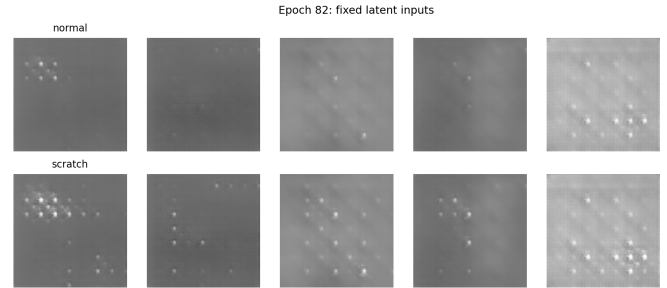
**Experiment 8: scratch-heavy exposure.** Increasing scratch exposure did not improve conditional control. Experiment 8 increased scratch exposure to fourteen of sixteen samples per batch, but coherent scratch geometry did not emerge. Later checkpoints instead contained repeated point-like, grid-like, and checkerboard artifacts. These structures appeared under both normal and scratch labels, and corresponding fixed-latent image pairs often remained visually similar despite the label change. The scratch-heavy regime therefore degraded normal-class fidelity without producing reliable scratch-class separation or geometric diversity.

The fixed-latent comparison was particularly informative because paired samples used the same latent vector and differed only in the requested class. Their visual similarity indicates that the generator was weakly responsive to the class label and often relied more heavily on the latent input and repeated architectural artifacts than on conditional semantic control. Training losses were monitored for instability, but the checkpoint images were treated as the primary evidence because adversarial losses do not directly measure conditional fidelity or defect geometry.

Overall, the staged framework ruled out two plausible explanations for the original failure. Simplifying the background and increasing scratch exposure did not yield reliable or diverse scratch generation. The evidence therefore points beyond data imbalance alone toward limitations in the present architecture and conditioning mechanism.



(a) Experiment 7 under balanced exposure. Only two of five scratch-conditioned samples show small defect-like cues, and geometric diversity remains limited.



(b) Experiment 8 under scratch-heavy exposure. Point-like artifacts appear under both labels, indicating weak conditional separation and degraded normal-class fidelity.

Figure 5: Final fixed-latent outputs from the controlled experiments. Simplifying the background produced only occasional scratch-like cues, while increasing scratch exposure introduced shared artifacts rather than stronger class control.

## 6 Discussion

The main contribution of this work is methodological. The project does not claim that a simple cDCGAN solves industrial synthetic defect generation. Instead, it demonstrates how a staged experimental framework can separate data-related explanations from limitations that persist under controlled conditions.

The original metal-surface experiments showed that the model learned shared surface texture more readily than defect-specific structure. Binary decomposition and targeted low-data interventions improved class separation, particularly for localized spot defects, but did not provide strong within-class diversity. These findings initially suggested that dominant background texture, limited defect visibility, and imbalance were major contributors to the weak scratch outputs.

The controlled follow-up experiments tested that interpretation directly. Experiment 7 removed the complex metal texture, balanced the classes, eliminated repetition-based oversampling, and exposed the model to known variation in scratch angle, location, length, and width. Despite these simplifications, reliable scratch generation did not emerge. Only a minority of final scratch-conditioned samples contained visible line-like cues, and those cues

remained short and repetitive. The result therefore weakens the explanation that the original failure was caused primarily by the metal background.

Experiment 8 tested a second plausible explanation: that equal exposure to the easier normal class suppressed learning of the more difficult scratch class. Increasing scratch exposure from eight to fourteen samples per batch did not improve the result. Instead, point-like and grid-like structures appeared under both labels, while normal-class quality deteriorated. Aggressive rebalancing therefore changed what the generator emphasized without producing useful conditional control.

The fixed-latent evaluation strengthens this conclusion. Corresponding normal and scratch outputs were generated from the same latent vectors, so the class label was the only changed input. Their visual similarity indicates that the generator did not consistently use the label to determine whether a scratch should be present. The model occasionally learned a weak defect-presence cue, but it did not learn a stable mapping from the scratch label to coherent, varied geometry.

These results do not establish that all conditional GANs are incapable of industrial scratch synthesis. The conclusion is specific to the present label-conditioned cDCGAN architecture, conditioning mechanism, loss formulation, and training setup. Nevertheless, the controlled evidence is strong enough to justify moving away from further count-ratio tuning or simple latent-dimension changes. The next phase should introduce mechanisms that explicitly represent defect geometry or operate by modifying an existing normal surface, including image-to-image translation [2, 7], defect-transfer and spatial-guidance methods [6], and richer conditional inputs that encode location, orientation, length, and width [4].

## 7 Conclusion

This report studied synthetic industrial surface defect generation under low-data constraints using a label-conditioned cDCGAN baseline. The original experiments showed that the model learned shared metal texture more readily than defect-specific structure, that localized spot defects were easier to model than elongated scratches, and that binary decomposition and targeted low-data interventions improved class consistency more clearly than diversity.

Two controlled follow-up experiments then tested whether the weak scratch results could be explained by complex background texture or insufficient scratch exposure. Removing the metal texture and balancing the classes produced only occasional, small, and repetitive scratch-like cues. Increasing scratch exposure to fourteen of sixteen samples per batch did not improve scratch geometry and instead introduced similar point- and grid-like artifacts under both labels. The fixed-latent comparisons further showed that the generator was weakly responsive

to the conditioning label.

These experiments support a narrower but stronger conclusion: background complexity and class imbalance were not sufficient explanations for the weak scratch results. Under the present architecture and training setup, the label-conditioned cDCGAN did not learn reliable class separation or controllable scratch geometry. Future work should therefore move beyond class-ratio tuning toward spatially guided, attribute-conditioned, defect-transfer, or image-to-image methods.

## 8 Ethics and Limitations Statement

This project is limited by both data scope and evaluation scope. The original dataset is small, imbalanced, and restricted to a single metal-surface family. The controlled follow-up dataset is procedural and intentionally simplified; it is useful for diagnosis but does not represent realistic industrial scratch appearance. Evaluation remains primarily qualitative and is based on checkpoint grids containing a limited number of fixed and randomly sampled latent inputs. Formal measures of scratch presence, orientation, length, spatial distribution, mode collapse, and conditional fidelity were not implemented in the present phase.

There are also broader risks in synthetic industrial data generation. Synthetic images may create a false sense of coverage if they are treated as replacements for real inspection data rather than controlled supplements. Poorly controlled generation may reinforce biases already present in the original dataset, overrepresent repetitive visual patterns, or produce unrealistic defect cues that could mislead downstream inspection models. For these reasons, synthetic data should be validated through task-based evaluation and explicit defect-geometry checks rather than visual plausibility alone.

## Future Work

The controlled experiments indicate that further tuning of class ratios within the same label-only cDCGAN is unlikely to provide the required geometric control. The next phase should therefore prioritize methods that make defect structure explicit.

- **Spatial or mask-guided defect synthesis:** provide an explicit defect mask or spatial condition so that location, orientation, extent, and shape are specified rather than inferred only from a global class label [6].
- **Image-to-image defect insertion:** investigate whether translation-based approaches can preserve a real normal-surface background while introducing a structured defect [2, 7].
- **Explicit geometric conditioning:** extend the conditional input beyond the class label to include nor-



malized angle, position, length, and width variables, allowing the requested scratch geometry to be evaluated directly [4].

- **Stronger quantitative evaluation:** measure scratch-presence rate, conditional-label fidelity, orientation and length distributions, image diversity, memorization, and downstream inspection-model performance [3].

## Code Availability

The codebase developed for this project is publicly available at: <https://github.com/SrujanVaddiparthi/synthetic-industrial-defect-generation>

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